A two-point exterior also fails at odd-degree equality

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be a transitive string C-group with connected diagram and all maximal omissions intransitive. Let $D = \{1, \dots, 6\}$. Assume the first three marks are an actual transitive head supported on D, and that

$$\tau = \rho_3 = (6\ b)(u\ v), \qquad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus \{1, \dots, 5\})$$

are literal permutation statements, where b, u, v are distinct points outside D. Put $\sigma = \rho_4$, $H = G_{\ge 5}$ and $K = G_{\ge 6}$. Assume σ fixes D and H fixes $D \cup \{b\}$. No transitivity assumption is made on $G_{\ge 4}$ or on the support of H.

Theorem 1. *If $r \ge 6$, $n = |\Omega| \in \{2r, 2r+1\}$, and every colour $j \ge 6$ has at least two full ambient-eligible edges, then the displayed configuration is impossible.*

Proof. Transitivity of R forces $x = b^\sigma \ne b$, because otherwise $R = \langle \tau, \sigma, H \rangle$ would preserve $\{6, b\}$. The pure alternating overlap argument gives

$$\mathrm{Alt}(D \cup \{b\}) \le G_{\le 3}, \qquad \mathrm{Alt}(D \cup \{b, x\}) \le G_{\le 4}.$$

For completeness, conjugating τ by the transitive head and multiplying two conjugates cancels $(u\ v)$, giving all pure three-cycles through b and two head points. Conjugating the resulting alternating group by σ and merging the two supports gives the second containment. If $x^\tau \ne x$, one more such overlap puts a pure A_4 on $\{6, b, x, x^\tau\}$ in

$$G_{\le 4} \cap R = \langle \tau, \sigma \rangle,$$

by full actual IP. A group generated by two involutions is dihedral, and its odd-order elements commute; that pure A_4 is impossible. Thus $\tau x = x$, so $x \notin \{u, v\}$. The same intersection contains a pure three-cycle q on $\{6, b, x\}$.

The set $F = \{u, v\}$ is preserved by H: the latter fixes D, b and commutes with τ. It cannot be preserved by σ, since then all of R would preserve it. If exactly one point of F moves outside, write $u^\sigma = w$ and $v^\sigma = v$. The two invariant triples $\{6, b, x\}$ and $\{u, v, w\}$ each carry an order-three restriction of $\tau\sigma$. Every element of odd order in its dihedral group is a power of that rotation. A power trivial on the second triple is divisible by three and trivial on the first. This contradicts the pure cycle q.

Both points therefore move outside F. Write $w = u^\sigma$, $z = v^\sigma$. The four points u, v, w, z are distinct and disjoint from $D \cup \{b, x\}$. The group K fixes D, b, x, preserves F, and preserves $F^\sigma = \{w, z\}$: it commutes with both τ and σ. Consequently it preserves

$$Y = \Omega \setminus (D \cup \{b, x, u, v, w, z\}), \qquad |Y| = n - 12.$$

All ambient-eligible edges of colours $j \ge 6$ lie in Y. Indeed the only possible edges outside are on F or F^σ. The first are joined by $\tau \in G_j$; the second by $\tau^\sigma = (6\ x)(w\ z) \in G_j$. Neither is eligible for the full omission of j.

If $r > 6$, apply the complete acquired localization theorem to the terminal interval $I = \{6, \dots, r-1\}$ and its proper invariant set Y. It counts every actual K-orbit in Y, including singletons, and gives

$$2(r-6) \le |Y| - o_K(Y).$$

For $n = 2r$ this is impossible, since the left side is $|Y|$. For $n = 2r + 1$, it forces $o_K(Y) = 1$: the nonempty set Y has odd size $2r - 11$ and K is transitive there. The involution σ preserves Y and centralizes K. An involution on an odd set has a fixed point, and a centralizing permutation fixing one point of a transitive action fixes every point. Hence $\sigma|_Y = 1$. If $r = 6$, then $|Y| = 0$ or 1, and this conclusion holds without the interval argument.

All points have now been accounted for. The permutation σ fixes D and Y and is exactly

$$(b\ x)(u\ w)(v\ z).$$

It is odd, contrary to $G \leq \mathrm{Alt}(\Omega)$. □

Attribution and scope. The pure three-cycle and confined-edge argument extends the accepted two-point exterior obstruction. Its full localization proof remains paid. The new odd-degree step uses the forced transitivity on Y to eliminate the last possible odd factor of σ. It uses no smaller-degree alternating rank induction and no new computation. The literal natural R and the exact two-point exterior remain hypotheses. This is one case of the needed odd-degree separator refinement, not a proof of that refinement or of the full L044 recovery. No whole-original, formal or question credit is claimed.